

33. Which of the following is the smallest muscle in the human body?

- (a) Bicep (b) Sartorius
(c) Stapedius (d) Tricep

M.P. P.C.S. (Pre) 2025

Ans. (c)

The stapedius muscle in the middle ear is the smallest muscle in the human body. The stapedius is located in the tympanic cavity of the middle ear. It connects the pyramidal eminence of the temporal bone to the neck of the stapes bone. Stapedius muscle stabilizes and controls the vibration of the stapes bone and it is one of the intratympanic muscles for the regulation of sound.

34. The strongest muscle in human body is found in :

- (a) Jaws (b) Thighs
(c) Neck (d) Hands

U.P.P.C.S. (Mains) 2008

Ans. (a)

The strongest muscle based on its weight is the masseter, known as jaw muscle. With all muscles of the jaw working together, it can close the teeth with a force as great as about 55 pounds (25 kilograms) on the incisors or about 200 pounds (90.7 kilograms) on the molars.

35. Match List-I with List-II and select the correct answer using the codes given below the lists :

List-I (Bones)	List-II (Names)
A. Breastbone	1. Clavicle
B. Collarbone	2. Patella
C. Knee-cap	3. Scapula
D. Shoulder Blade	4. Sternum

Code :

	A	B	C	D
(a)	4	1	3	2
(b)	1	4	2	3
(c)	1	4	2	3
(d)	4	1	2	3

I.A.S. (Pre) 2001

R.A.S./R.T.S. (Pre) 2013

Ans. (d)

The correctly matched lists are as follows :

Breastbone	—	Sternum
Collarbone	—	Clavicle
Knee-cap	—	Patella
Shoulder Blade	—	Scapula

II. Digestive System

Notes

Digestion :

- Digestion is the breakdown of large insoluble food molecules into small water-soluble food molecules so that they can be absorbed into the watery blood plasma. Digestion is a form of catabolism that is often divided into two processes based on how food is broken down : mechanical and chemical digestion.
- Mechanical digestion** refers to the physical breakdown of large pieces of food into smaller pieces which can subsequently be accessed by digestive enzymes.
- In **chemical digestion**, enzymes break down food into the small molecules the body can use.

Human digestive system :

- The human digestive system consists of the alimentary canal (gastrointestinal tract) plus the accessory organs of digestion (the tongue, salivary glands, pancreas, liver and gall bladder). The alimentary canal is a long tube through which food is taken into the body and digested.
- In human, this passage is about 30 feet (9 metre) long from the oesophagus to the anus. The **buccopharyngeal cavity, oesophagus, stomach** and **intestines** are the main parts of the alimentary canal.

Digestion in Mouth :

- Saliva** is the watery and usually somewhat frothy substance produced in the mouth by salivary glands.
- Human saliva comprises about 99.5% water plus electrolytes, mucus, white blood cells, epithelial cells (from which DNA can be extracted), enzymes such as amylase (ptyalin) and lipase, antimicrobial agents such as secretory IgA, and lysozymes.
- The amount of saliva that is produced in a healthy person per day ranges from about 0.75 to 1.5 litres.
- The digestive functions of saliva include moistening food, and helping to create a food bolus, so it can be swallowed easily.
- Amylase enzyme (ptyalin) breaks some starches into maltose and dextrin.
- The enzyme lysozyme hydrolyzes the cell wall of bacteria resulting in their killing.

In human, digestion starts from the mouth

Oesophagus :

- Oesophagus or esophagus is also called the gullet or food pipe.
- It is a muscular tube about 25 cm. long. It connects mouth to the stomach.

- When the food is swallowed, the walls of the oesophagus squeeze together (contract). This moves the food down the oesophagus to the stomach.

Digestion in Stomach :

- The stomach is a muscular sac that lies between the oesophagus and small intestine, in the upper abdomen.
- The stomach is with strong muscular walls.
- In addition to holding the food, it is also a mixer and grinder.
- The stomach secretes acid and powerful enzymes that continue the process of breaking down of the food.
- Substances secreted by gastric glands of the stomach and their functions are as follows :

(i) Hydrochloric acid (HCl) :

- Hydrochloric acid provides the acidic medium to the food for digestion.
- It kills the bacteria.
- It dissolves the foreign matter such as small stones, or bones that come along with food.
- It activates the enzyme pepsin and renin.

(ii) Mucin :

- Mucin is a **glycoprotein**. It forms a thin layer over the internal layer of the stomach to protect it from being damaged by hydrochloric acid.

(iii) Pepsin and Renin :

- Pepsin and Renin are protein digestive enzymes.
- It breaks down the protein into peptone and polypeptides.

(iv) Lipase :

- It breaks down fat into glycerol and fatty acids.

Liver :

- The liver is the largest gland of the human body.
- The liver is an essential organ that has many functions in the body, including making proteins and blood clotting factors, manufacturing triglycerides and cholesterol, glycogen synthesis, bile production, the synthesis of urea, breakdown of R.B.C. etc.

Bile :

- Bile or gall is a dark green to yellowish brown fluid produced by the liver, that aids the digestion of lipids in the duodenum.
- The composition of hepatic bile is 97% water, 0.7% bile salts, 0.2% bilirubin, 0.51% fats (cholesterol, fatty acids and lecithin) and little amount of inorganic salts.
- About 800 to 1000 ml of bile is produced per day in adult human beings.
- Bile or gall acts to some extent as a surfactant, helping to emulsify the fats/lipids in food.

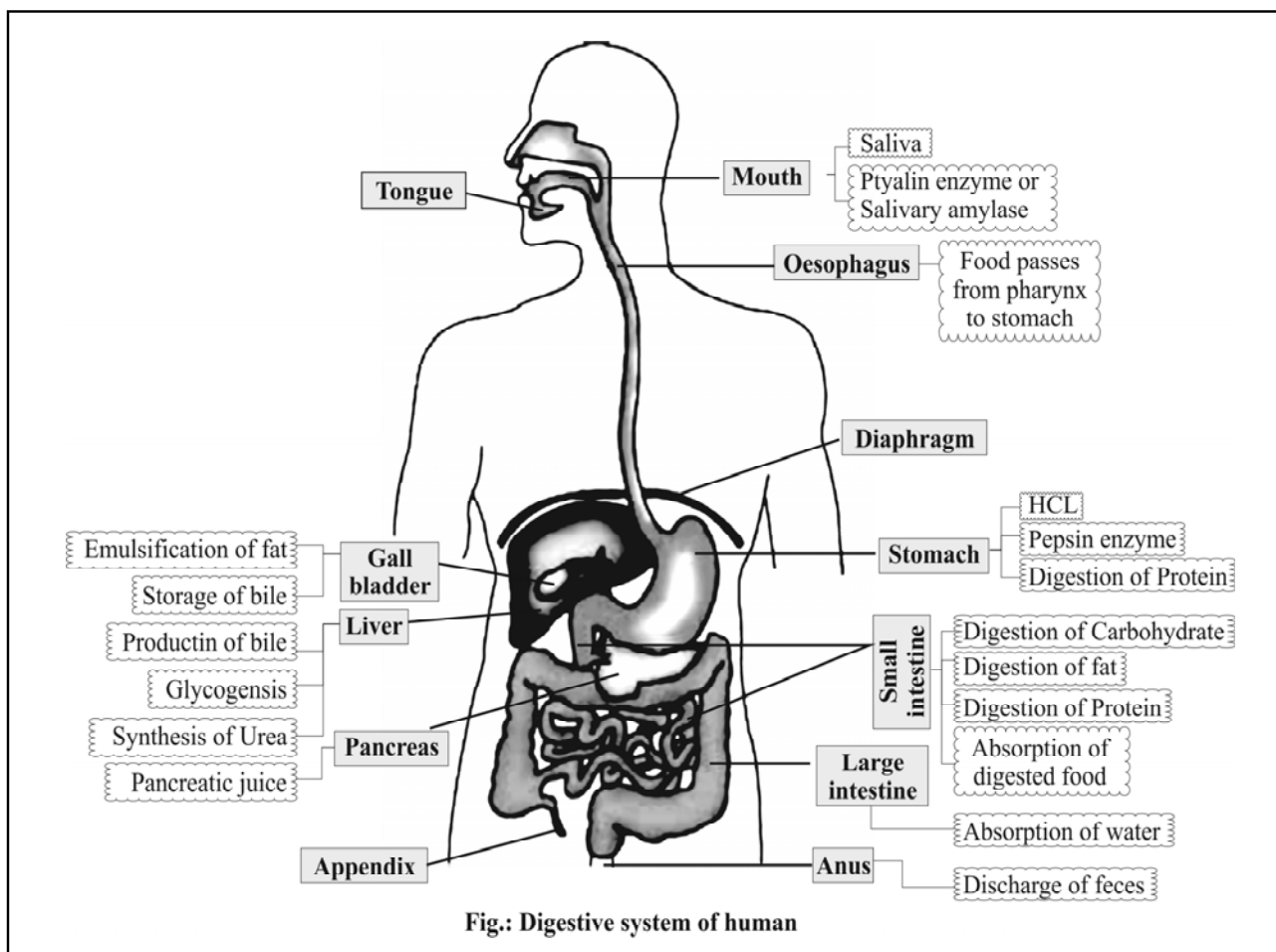


Fig.: Digestive system of human

Pancreas :

- The pancreas is an organ of the digestive system and endocrine system of vertebrates. It is both an endocrine gland and exocrine gland. Hence it is called **the mixed gland**. In adults, it is about 12-15 cm long, lobulated and salmon-coloured in appearance.
- It is the second one of the largest gland in human.
- In human, it is located in the abdomen behind the stomach.
- As an endocrine gland, it functions mostly to regulate blood sugar levels, secreting the hormones **insulin**, **glucagon** and **somatostatin**, and pancreatic polypeptide.

Pancreatic juice :

- Pancreatic juice is a liquid secreted by the pancreas, which contains a variety of enzymes.
- The enzymes present in pancreatic juice are as follows :
 - (i) **Amylase**— It converts starch into disaccharides.
 - (ii) **Trypsin**— It converts protein into smaller peptides which are then further hydrolyzed into amino acids.
 - (iii) **Lipase**— It converts fat into glycerol and fatty acids.
 - (iv) **Elastase**— It break down proteins.
 - (v) **Nuclease**— It breaks down the nucleic acid into nucleotide.
- Since all types of food are digested by pancreatic juice, so it is called **complete digestive juice**.

Digestion of food in duodenum takes place in alkaline medium

Small Intestine :

- The small intestine is a long, narrow, folded or coiled tube extending from the stomach to the large intestine; it is the region where most digestion and absorption of food takes place.
- It is about 6.7 to 7.6 metres (22 to 25 feet) long in adults, highly convoluted and contained in the central and lower abdomen.
- Three successive regions of the small intestine are distinguished : **duodenum**, **jejunum** and **ileum** whose length is about 25 cm., 2.5 metre and 3.5 metre respectively.

Digestion in Duodenum :

- The first part of small intestine, the duodenum, is adjacent to the stomach.
- Ducts from the liver, gallbladder and pancreas, enter the duodenum to provide the juices that neutralize acids coming from the stomach and help to digest proteins, carbohydrates and fats.

Digestion in Jejunum :

- The jejunum is the second part of the small intestine.
- Its lining is specialized for the absorption by enterocytes or small nutrient molecules which have been previously digested by enzymes in the duodenum.
- Once absorbed, nutrients (with the exception of fat, which goes to the lymph) pass from the enterohepatic circulation and enter the liver, where the blood is processed.

- The jejunum is involved in magnesium absorption.

Digestion in Ileum :

- The ileum is the lower part of the small intestine and the largest section of the alimentary canal.
- The inner wall of the ileum contains secretory cells, some of which secrete **mucus** and other **secrete** an alkaline fluid known as internal juice **succus entericus**.
- The arrival of chyme in the ileum stimulates the secretion of intestinal juice.
- The juice contains the following enzymes that carry out digestive functions which are as follows—
 - (i) **Erepsin**—breaks down peptones into amino acids.
 - (ii) **Lipase**—breaks down fats into fatty acid and glycerol.
 - (iii) **Maltase**—breaks down maltose into glucose.
 - (iv) **Sucrase**—speeds up the breakdown of sucrose into glucose and fructose.
 - (v) **Nucleotidase**—It breaks down the nucleotide into nitrogen base, sugar and phosphoric acid.

The process of digestion is completed in the ileum

- The resulting watery substance of food is called chyle and contains soluble products of digestion ready to be absorbed.

Absorption of digested food :

- Absorption is the transfer of digested food to the blood or lymph through the mucosa of the ileum.
- The inner epithelial lining of the ileum is folded to form **villi**.
- **Villi** are many folds in the inner wall of the ileum which causes an increase in the surface area of **the small intestine** that is subjected to the absorption of digested food.

Large Intestine (Colon) and Defecation :

- The undigested food passes to the large intestine.
- Water and salts from the undigested food are absorbed through the lining of the epithelial wall of the large intestine, as it has many convolutions to help in the absorption.
- Wastes become **semisolid faeces** and have a bad odour, due to the presence of certain types of bacteria. These bacteria decompose the undigested food producing different gases like methane, sulphur dioxide, ammonia, carbon dioxide, etc.

Caecum and Vermiform Appendix :

- The appendix is a finger-like, blind-ended tube connected to the caecum.
- The caecum is a pouch-like structure of the colon, located at the junction of the small intestine and the large intestine.
- The main functions of the caecum are to absorb fluids and salts that remain after completion of intestinal digestion and absorption, and to mix its contents with a lubricating substance, mucus.
- In human vermiform appendix is a **vestigial remnant**, that is, it was once useful to humans in the digestion of

food but as human has evolved, it has lost its functions and become redundant.

- In **herbivores**, the vermiform appendix is well developed and helps in the digestion of **cellulose**.

Enzyme :

- Enzyme is basically protein.
- Enzyme is a substance produced by a living organism which acts as a catalyst to bring about a specific biochemical reaction.
- **Zymase** enzyme converts glucose into ethyl alcohol and carbon dioxide. It occurs naturally in **yeasts**.

Question Bank

1. In human body most of the digestive process, takes place in –

- (a) Pancreas (b) Large intestine
(c) Small intestine (d) Stomach

U.P.P.C.S. (Pre) 1991

Ans. (c)

Digestion begins in the mouth when we chew and swallow and is completed in the small intestine. The small intestine is a long tube loosely coiled in the abdomen (spread out, it would be more than 20 feet long). The small intestine continues the process of breaking down food by using enzymes released by the pancreas and bile from the liver.

2. Which of the following are not the tastes of the tongue?

1. Sweet 2. Bitter
3. Salty 4. Spicy
5. Umami 6. Sour
7. Pungent

Select the correct answer using the codes given below.

- (a) 2, 5 and 7 (b) 1, 3 and 4
(c) 4 and 7 (d) 3 and 6

69th B.P.S.C. (Pre) 2023

Ans. (c)

Tongue of human being has different types of papillae associated with taste buds and are responsible for detecting the tastes like – Sweet, Bitter, Salty, Umami and Sour. They do not detect the 'Spicy' taste and the 'Pungent' which is associated with smell. The sensation of something as 'hot' or 'spicy' is quite often described as a taste. Technically, this is just a pain signal sent by the nerves that transmit touch and temperature sensations. The substance 'capsaicin' in foods seasoned with chilli causes a sensation of pain and heat. Hence, option (c) is the correct answer.

3. The sensitive area of the human tongue to bitterness is:

- (a) Tip (b) Middle part
(c) Posterior part (d) Edge

- (e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (c)

Sweet, sour, salty, bitter and savory (umami) tastes can actually be sensed by all parts of the human tongue. Only the sides of the tongue are more sensitive than the middle overall. This is true for all tastes with one exception : the back of our tongue (i.e. posterior part) is very sensitive to bitter tastes. This is apparently to protect us so that we can spit out spoiled or poisonous foods or substances before they enter the throat and are swallowed.

4. In the buccal cavity of human being digestion of which one of the following get started?

- (a) Protein (b) Fat
(c) Carbohydrate (d) None of the above

M.P.P.C.S. (Pre) 2019

Ans. (c)

In the buccal cavity of human being amylase enzyme in saliva breaks down starch (a form of carbohydrate) into maltose and dextrin. Thus, in the buccal cavity digestion of carbohydrate gets started.

5. The saliva helps in the digestion of –

- (a) Starch (b) Proteins
(c) Fibres (d) Fat

R.A.S./R.T.S.(Pre) 2012

Ans. (a)

Saliva is a watery substance located in the mouths of man and animals, secreted by the salivary glands. Human saliva is about 99.5% water, while the other 0.5% consists of electrolytes, mucus, glycoproteins, enzymes (eg. amylase) and antibacterial compounds such as secretory IgA and lysozyme. The enzymes found in saliva are essential to begin the process of digestion of dietary starches.

6. Which of the following is digested with the help of saliva?

- (a) Protein (b) Starch
(c) Fibre (d) None of the above

Jharkhand P.C.S. (Pre) 2013

Ans. (b)

See the explanation of above question.

7. Enzyme involved in the hydrolysis of starch to glucose is-

- (a) Invertase (b) Amylase
(c) Dehydrogenase (d) Anhydrase

R.A.S./R.T.S. (Pre) 1999

Ans. (b)

Amylase is present in the saliva of humans and some other mammals, where it begins the chemical process of digestion. Amylase is an enzyme that catalyzes the hydrolysis of starch into glucose.

8. Assertion (A) : In human body, the liver has an important role in fat digestion.

Reason (R) : Liver produces two important fat-digesting enzymes.

Code :

- (a) Both (A) and (R) are individually true and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are individually true, but (R) is not the correct explanation of (A).
- (c) (A) is true, but (R) is false.
- (d) (A) is false, but (R) is true.

I.A.S. (Pre) 2008

Ans. (c)

The liver does not produce fat-digesting enzymes. It detoxifies chemical and metabolizes drugs. The liver has multiple functions such as secretion of bile, storage of fat, synthesis of urea, glycogenesis, gluconeogenesis, and glycogenolysis etc. Although the bile lacks digestive enzymes, yet it plays an important part in the digestion, especially in digestion of fats by bringing about their emulsification.

9. In which organ hydrochloric acid is secreted?

- (a) Intestine
- (b) Liver
- (c) Stomach
- (d) Pancreas

M.P. P.C.S. (Pre) 2021

Ans. (c)

Hydrochloric acid is secreted by the parietal cells (also known as oxyntic cells) of the gastric glands in the stomach. Gastric juice contains mucous, hydrochloric acid and certain enzymes such as pepsin. Hydrochloric acid kills the harmful bacteria present in the food. Hydrochloric acid secretion by the gastric parietal cells is necessary for pepsinogen activation and the reduction of bacterial colonization.

10. Which of the following acid is secreted in the stomach ?

- (a) Sulphuric acid
- (b) Nitric acid
- (c) Phosphoric acid
- (d) Hydrochloric acid

70th B.P.S.C. (Pre) 2024

Ans. (d)

See the explanation of above question.

11. Enzyme is a –

- (a) Vitamin
- (b) Bacterium

(c) Bio-catalyst

(d) Virus

Uttarakhand P.C.S. (Pre) 2010

Ans. (c)

Enzymes are biological molecules (proteins) that act as bio-catalysts and helps complex reactions occur in the body. These are basically proteins which accelerate chemical reactions in living systems. Almost all metabolic processes in the cell need enzymes catalysis in order to occur at rates fast enough to sustain life.

12. Enzymes are basically –

- (a) Fats
- (b) Sugars
- (c) Proteins
- (d) Vitamins

U.P. Lower Sub. (Mains) 2013

U.P.P.C.S. (Spl.) (Mains) 2004

Uttarakhand P.C.S. (Mains) 2002

Uttarakhand P.C.S. (Pre) 2002

M.P.P.C.S. (Pre) 1990

U.P.P.C.S. (Pre) 1996

Ans. (c)

See the explanation of above question.

13. Most of the enzymes are :

- (a) proteins
- (b) lipids
- (c) acids
- (d) alkalis
- (e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (a)

See the explanation of above question.

14. Enzymes are –

- (a) Fats
- (b) RNA
- (c) Proteins
- (d) None of the above

Jharkhand P.C.S. (Pre) 2013

Ans. (c)

See the explanation of above question.

15. Substances which are responsible for catalyzing chemical reactions in living systems are :

- (a) Bacteria
- (b) DNA
- (c) Enzymes
- (d) Proteins

44th B.P.S.C. (Pre) 2000

Ans. (c)

See the explanation of above question.

16. Which of the following statements about enzymes is/ are correct?

1. They are biocatalysts.
2. They perform their action on the site where they are produced.

Select the correct answer from the codes given below :

Codes :

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

U.P.P.C.S. (Pre) 2019

Ans. (c)

Enzymes can be defined as biological polymers that catalyze biochemical reactions. Thus, they are biocatalysts and accelerate chemical reactions in the human body. Generally, enzymes perform their action on the site where they are produced. Hence, both statements are correct.

17. Which enzyme can catalyze the conversion of glucose to ethanol?

- (a) Zymase (b) Invertase
(c) Maltase (d) Diastase

Jharkhand P.C.S. (Pre) 2013

Ans. (a)

Zymase is an enzyme complex that catalyzes the fermentation of sugar into ethanol and carbon dioxide. It occurs naturally in yeasts. Zymase activity varies among yeast strains.

18. Production of which one of the following is a function of the liver?

- (a) Lipase (b) Urea
(c) Mucus (d) Hydrochloric acid

I.A.S. (Pre) 2007

Ans. (b)

The liver is the largest gland in human body. It is located in the abdominal cavity, just below the diaphragm and has two lobes. It is an important digestive organ in human body. Its main function is the production of bile, storage of fat and conversion of harmful ammonia to urea (urea is one of the end products of protein metabolism that is excreted in the urine).

19. Digestion of Lipids takes place in the presence of some of the following :

1. Bile acids 2. Lipase
3. Pepsin
of these,

- (a) Only 1 is correct
(b) Only 1 and 3 are correct
(c) Only 1 and 2 are correct
(d) 1, 2 and 3 are correct

U.P. P.C.S. (Mains) 2016

Ans. (c)

Most of the lipids in our food are present in the form of simple fats or triglycerides. Lipase enzyme plays an important role in the digestion of Lipids. Lipase breaks down lipids into fatty acids and monoglycerides. Bile probably contains no digestive enzyme, yet it plays an important role in the digestion and absorption of fat. It facilitates emulsification of fats. Pepsin is a protein digesting or proteolytic enzyme.

20. The source of the enzyme 'lipase' is :

- (a) Kidney (b) Pancreas
(c) Liver (d) Heart
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (b)

The enzyme 'lipase' is produced in the pancreas, mouth and stomach. It is a fat hydrolysing enzyme. Its function is to digest fats and lipids, helping to maintain correct gallbladder function.

21. Which one of the following is not a digestive enzyme in the human system?

- (a) Trypsin (b) Gastrin
(c) Ptyalin (d) Pepsin

I.A.S. (Pre) 2007

Ans. (b)

Trypsin, ptyalin, and pepsin are digestive enzymes whose secretion is performed by pancreas, salivary glands and stomach respectively. Gastrin is a peptide hormone that stimulates secretion of gastric acid (HCl) by the parietal cells of the stomach and aids in gastric motility. It is released by G cells in the pyloric antrum of the stomach, duodenum, and the pancreas.

22. In human body, the digestion of protein begins in which of the following organs?

- (a) Liver (b) Mouth
(c) Small intestine (d) Stomach

U.P. P.C.S. (Pre) 2020

Ans. (d)

In human body, the digestion of protein begins in the stomach with the action of enzyme pepsin. Pepsin is secreted by the gastric glands of the stomach which digests the protein into smaller peptides and amino acids, which can be readily absorbed by the small intestine.

23. Which one of the following enzymes convert proteins into amino acids?

- (a) Pepsin (b) Lactase

(c) Urease

(d) Zymase

U.P. R.O./A.R.O. (Pre) 2017

Ans. (a)

Pepsin is a protein digestive enzyme which is secreted by gastric glands of the stomach which digests the protein into smaller peptides and amino acids, which can be readily absorbed by the small intestine. Lactase is milk digesting enzyme which converts lactose in glucose and galactose. Urease enzyme catalyzes the hydrolysis of urea forming carbon dioxide and ammonia. Zymase enzyme occurs naturally in yeast that catalyzes the fermentation of sugar into ethanol and carbon dioxide.

24. Assertion (A) : All the proteins in our food are digested in small intestine only.

Reason (R) : The protein-digesting enzyme from the pancreas are released into the small intestine.

Code :

- (a) Both (A) and (R) are individually true and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are individually true, but (R) is not a correct explanation of (A).
- (c) (A) is true, but (R) is false.
- (d) (A) is false, but (R) is true.

I.A.S. (Pre) 2005

Ans. (d)

Assertion (A) is false, but reason (R) is true. Protein digestion begins in the stomach with the action of an enzyme called pepsin. Pepsin is the active protein-digesting enzyme of the stomach. Protein-digesting enzyme trypsin is produced in an inactivate form in the pancreas and is activated in the small intestine, where it digests protein. Duodenum is the first and shortest segment of the small intestine. In the duodenum, trypsin catalyzes the hydrolysis of peptide bonds, breaking down proteins into smaller peptides. Enzymes continue to break down polypeptides and peptides into amino acids. As amino acids are very small, they are able to be absorbed through the small intestine lining and into the bloodstream.

25. An enzyme which helps in the digestion of protein is :

- (a) Urease
- (b) Sulfatase
- (c) Trypsin
- (d) Protease

U.P.P.C.S (Pre) 2010

Ans. (c)

See the explanation of above question.

26. Which of the following changes is catalyzed by the enzyme trypsin?

- (a) Proteins to peptones
- (b) Proteins to peptides
- (c) Proteins to amino acids
- (d) Starch to glucose

R.A.S./R.T.S. (Pre) 1999

Ans. (c)

In the duodenum, trypsin catalyzes the hydrolysis of peptide bonds, breaking down proteins into smaller peptides. The peptide products are then further hydrolyzed into amino acids via other proteases, rendering them available for absorption into the bloodstream.

27. In the human body, which structure is the appendix attached to?

- (a) The large intestine
- (b) The small intestine
- (c) The gall bladder
- (d) The stomach

I.A.S. (Pre) 2007

Ans. (a)

The appendix is a blind-ended tube connected to the caecum, that is considered to be the beginning of the large intestine from which it develops embryologically. The caecum is a pouch-like structure of the colon, located at the junction of the small and the large intestines.

28. Which one of the following is the correct sequence in the order of decreasing length of the three structural part given below of small intestine in the human body?

- (a) Jejunum – Duodenum – Ileum
- (b) Ileum – Duodenum – Jejunum
- (c) Jejunum – Ileum – Duodenum
- (d) Ileum – Jejunum – Duodenum

I.A.S. (Pre) 2007

Ans. (d)

The small intestine is divided into three structural parts. The duodenum is a short structure (about 20–25 cm long) continuous with the stomach and shaped like 'C'. It surrounds the head of the pancreas. The duodenum contains Brunner's glands, which produce a mucus-rich alkaline secretion containing bicarbonate. These secretions, in combination with bicarbonate from the pancreas, neutralizes the stomach acids contained in the gastric chyme. The jejunum is the midsection of the small intestine, connecting the duodenum to the ileum. It is about 2.5m long and contains the plicae circulares and villi that increases its surface area. The ileum is the final section of the small intestine. It is about 3.5 m long and contains villi similar to the jejunum. Hence, option (d) is the correct answer.

29. Given below are two statements :

Assertion (A) : Human body is incapable of digesting cellulose.

Reason (R) : Starch degrading enzyme, diastase, is found in human body.

Choose the correct answer from the code given below:

Codes :

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (b) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (c) (A) is true but (R) is false.
- (d) (A) is false but (R) is true.

U.P.P.C.S. (Pre) 2017

Ans. (b)

In human digestive system, enzymes that can digest cellulose are not found, but starch digestive enzymes, diastase which break starch into maltose are found in human saliva. Thus both the statements are correct, but the cellulose can not be digested due to diastase. So, the second statement does not explain the first statement.

30. Identify the correct pair of enzyme and its function :

- (a) Acid phosphatase – Mitochondrial marker enzymes related to oxidative functions.
- (b) E. coli restriction endonuclease-II – cuts DNA at specific places.
- (c) Acrosin – cortical granules of ovum secrete it to prevent polyspermic fertilization.
- (d) Succinic dehydrogenase – Lysosomal marker enzyme, hydrolyses metabolites.

R.A.S./R.T.S. (Re. Exam) (Pre) 2013

Ans. (b)

Restriction endonuclease Eco R II (pronounced 'eco R two'), an enzyme of restriction modification system (RM) naturally found in Escherichia coli, are a special class of enzyme that recognize and cleave (cuts) DNA at specific places. They produce small well-defined fragments of DNA that help to characterize genes and genomes and that produce recombinant DNAs.

31. Match List-I with List-II and select the correct answer using the codes given below the lists :

List-I	List-II
A. Ptyalin	1. Converts angiotensinogen in blood into angiotensin
B. Pepsin	2. Digests starch
C. Renin	3. Digests proteins
D. Oxytocin	4. Hydrolyses fats
	5. Induces contraction of smooth muscles

Code :

	A	B	C	D
(a)	2	3	1	5
(b)	3	4	2	5
(c)	2	3	5	1
(d)	3	1	2	4

I.A.S. (Pre) 2001

Ans. (a)

Ptyalin (Amylase) – An enzyme in the saliva that converts starch into dextrin and maltose.

Pepsin – An enzyme that splits proteins into proteoses and peptones.

Renin – A proteolytic enzyme secreted by the kidneys that convert angiotensinogen into angiotensin.

Oxytocin – A polypeptide hormone, produced by the posterior lobe of the pituitary gland that stimulates contraction of the smooth muscle of the uterus.

Hence, option (a) is the correct answer.

32. Match List-I with List-II and select the correct answer by using the codes given below :

List I	List II
A. Vitamin	1. Pepsin
B. Enzyme	2. Carotene
C. Hormone	3. Keratin
D. Protein	4. Testosterone/Progesterone

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	2	1	4	3
(c)	2	1	3	4
(d)	1	2	4	3

U.P. R.O./A.R.O. (Mains) 2021

U.P. Lower Sub. (Pre) 1998

I.A.S. (Pre) 1996

Ans. (b)

The correctly matched lists are as follows :

Vitamin	–	Carotene
Enzyme	–	Pepsin
Hormone	–	Testosterone/Progesterone
Protein	–	Keratin

33. The ultimate substance to which the carbohydrates are degraded, is :

- (a) amino acid
- (b) glycerol
- (c) glucose
- (d) maltose
- (e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (c)

Carbohydrates in the diet provide the major exogenous source for glucose, which is the primary energy source for cells. The goal of carbohydrate digestion is to break down all disaccharides and complex carbohydrates into monosaccharides for absorption. During digestion, starches and sugars are broken down both mechanically (e.g. through chewing) and chemically (e.g. by enzymes) into the single units of glucose, fructose, and/or galactose, which are absorbed into the blood stream and transported for use as energy throughout the body.

34. Extra glucose in body gets converted into glycogen and is stored in

- (a) Stomach
- (b) Liver
- (c) Pancreas
- (d) Bile

U.P.P.C.S. (Mains) 2004

Ans. (b)

The liver is the largest gland of the human body. Extra glucose in the body gets converted into glycogen and is stored in the liver. Glucose is a simple sugar found in carbohydrates. Once carbohydrates are absorbed from food, they are carried to the liver for processing.

35. Energy is stored in liver and muscles in the form of :

- (a) Carbohydrate
- (b) Fat
- (c) Protein
- (d) Glycogen

56th to 59th B.P.S.C. (Pre) 2015

Ans. (d)

Glycogen is the principal storage form of glucose or energy. In humans, it is stored in liver and muscle cells and can be converted to glucose, if needed.

36. How is liver affected on fasting for more than 10 days?

- (a) Glucose level in liver diminished
- (b) Glucose level in liver increases

(c) In liver triglycerides decrease

(d) In liver triglycerides increase

U.P. R.O./A.R.O. (Pre) 2021

Ans. (a)

Glucose level in liver diminished on fasting for more than 10 days. Researchers found that during fasting, the liver is busy producing glucose from its stored glycogen. In the absence of dietary sugars and carbohydrates, glucose is obtained from the breakdown of stored glycogen. Glycogen is a readily-accessible storage form of glucose, stored in notable quantities in the liver and skeletal muscle. When the glycogen reserve is depleted, glucose can be obtained from the breakdown of fats from adipose tissue. Fats are broken down into glycerol and free fatty acids, with the glycerol being turned into glucose in the liver via the gluconeogenesis pathway. When even the glucose made from glycerol reserves start declining, the liver starts producing ketone bodies. Ketone bodies are short-chain derivatives of the free fatty acids, and can cross the blood-brain barrier, meaning they can be used by the brain as an alternative metabolic fuel. After 2 or 3 days of fasting, the liver begins to synthesize ketone bodies from precursors obtained from fatty acid breakdown. During starvation, following depletion of hepatic glycogen, amino acids become the major source for glucose homeostasis. After several days of fasting, all cells in the body begin to break down protein. This releases amino acids into the bloodstream, which can be converted into glucose by the liver. As starvation progresses, gluconeogenesis diminishes in the liver but increases in the kidney as the need for ammonia excretion increases.

37. The stones present in the human gallbladder primarily affect the digestion of

- (a) Fats
- (b) Proteins
- (c) Carbohydrates
- (d) Nucleic acids

U.P. P.C.S. (Mains) 2017

U.P.P.C.S. (Pre) 2017

Ans. (a)

Gall stones are hard pebble-like pieces of material, usually made of cholesterol and bilirubin, that develop in the gallbladder. When gall stones block the bile duct, the bile secreted by the liver does not reach to the duodenum. The bile is primarily responsible for the digestion of fat. Thus, the stones present in the gallbladder primarily affect the digestion of fat.

38. Which is the largest gland in the human body?

- (a) Liver
- (b) Pancreas
- (c) Thyroid gland
- (d) Stomach

M.P.P.C.S. (Pre) 1990

Ans. (a)

The liver is the largest internal organ and the largest gland in the human body. The liver is about 22 cm long and 15 cm wide reddish brown gland, weighing around 1.5 kg. It plays an active role in the process of digestion of fats, carbohydrate through the production of bile.

39. The largest gland in human body is –

- (a) Pancreas (b) Liver
(c) Salivary gland (d) Adrenal gland

M.P. P.C.S. (Pre) 2016

Ans. (b)

See the explanation of the above question.

40. Bile is stored in which of the following organ?

- (a) Spleen (b) Pancreas
(c) Appendix (d) Gallbladder

R.A.S./R.T.S. (Pre) 1992

Ans. (d)

Bile is a fluid that is made and released by the liver and stored in the gallbladder. It is stored and concentrated in the gallbladder unit which is needed to digest food. In adults, the gallbladder measures approximately 8 centimetres in length and 4 centimetres in diameter.

41. Bile is produced in which part of the body?

- (a) Liver (b) Spleen
(c) Gallbladder (d) Pancreas

U.P.P.C.S. (Pre) 1999

Ans. (a)

See the explanation of above question.

42. Which part of alimentary canal receives bile from the liver ?

- (a) Stomach (b) Oesophagus
(c) Small intestine (d) Large intestine

70th B.P.S.C. (Pre) 2024

Ans. (c)

The small intestine, specifically the duodenum (the first part of the small intestine), is the part of the alimentary canal that receives bile from the liver, which is stored and concentrated in the gallbladder. The liver produces bile, a fluid that aids in the digestion and absorption of fats. Bile is stored and concentrated in the gallbladder. Bile flows from the liver and gallbladder through a network of ducts, including the common hepatic duct and the cystic duct, ultimately forming the common bile duct. The common bile duct empties into

the duodenum, the first part of the small intestine. In the duodenum, bile emulsifies fats, breaking them down into smaller droplets, which allows digestive enzymes to more effectively break down the fats.

43. Consider the following :

- i. This is the second largest gland of the human body.
ii. Its secretion comes out from a characteristic type of cells and is alkaline.
iii. Its secretion is said to be complete digestive juice.

These characteristics are of

- (a) liver (b) spleen
(c) pancreas (d) salivary glands

U.P.P.C.S. (Spl) (Mains) 2008

Ans. (c)

The liver is the largest gland while pancreas is the second largest gland of the human body. The pancreas is a glandular organ in the digestive system and endocrine system of the vertebrates. In humans, it is located in the abdominal cavity behind the stomach. The pancreas is also a digestive organ, secreting pancreatic juice that is alkaline. Its secretion is said to be complete digestive juice.

44. With reference to normal human beings, consider the following statements :

1. In response to the presence of HCl, secretin is produced from the duodenum.
2. Enterogastrone is produced in the small intestine in response to the presence of fatty acids.

Which of these statements is / are correct ?

- (a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2003

Ans. (c)

Secretin is synthesized in cytoplasmic secretory granules of S-cells, which are found mainly in the mucosa of the duodenum. It is released into circulation and/or intestinal lumen in response to acidity (low duodenal pH) which is due to hydrochloric acid (HCl) in the chyme that enters the duodenum from the stomach. Its primary function is to neutralize the pH in the duodenum, allowing digestive enzymes from the pancreas to function optimally. Enterogastrone is a hormone secreted by the mucosa of the duodenum (first section of the small intestine) in the lower gastrointestinal tract when fatty acids are in the stomach or small intestine. It inhibits the caudal (forward, analward) motion of the contents of chyme.